ABV IIITM STUDENT BRANCH

IEEE MP SECTION

HACKSAGON

A NATIONAL LEVEL SOFTWARE & HARDWARE HACKATHON

Livestock Monitoring System

Team - Alpha

SOFTWARE +HARDWARE

The system integrates ***IoT sensors, AI-based Analytics, Cloud computing***

to provide real-time monitoring of livestock health.

Wearable sensors collect physiological data, which is processed through

AI algorithms & displayed on a user-friendly mobile/web dashboard.

PROBLEM STATEMENT

Farmers often detect diseases too late due to a **lack of real-time health monitoring**, leading to animal deaths and significant financial losses as treatment is delayed.

STATISTCS

➢ Cattle and Buffaloes

Lumpy Skin Disease (LSD):
Deaths (2022): Over 97,000 cattle
Infections: More than 2 million

Foot and Mouth Disease (FMD):
Annual Loss: ₹20,000 crore ($2.4 billion)

Haemorrhagic Septicaemia (HS):
Deaths: 58% to 77% of bovine deaths in India
Mortality Rate: Animals can die within 24 hours if untreated

➢ Poultry (Chickens, Ducks, etc.):

Avian Influenza (Bird Flu):
Culling (2021): Over 1.6 million poultry birds
Newcastle Disease:
Mortality Rate: 90% in unvaccinated birds

➢ Sheep and Goats:

Peste des Petits Ruminants (PPR):
Mortality Rate: 70-80% of infected animals if untreated
Annual Loss: ₹5,000 crore ($600 million)
Blue Tongue Disease: Causes high mortality in sheep.

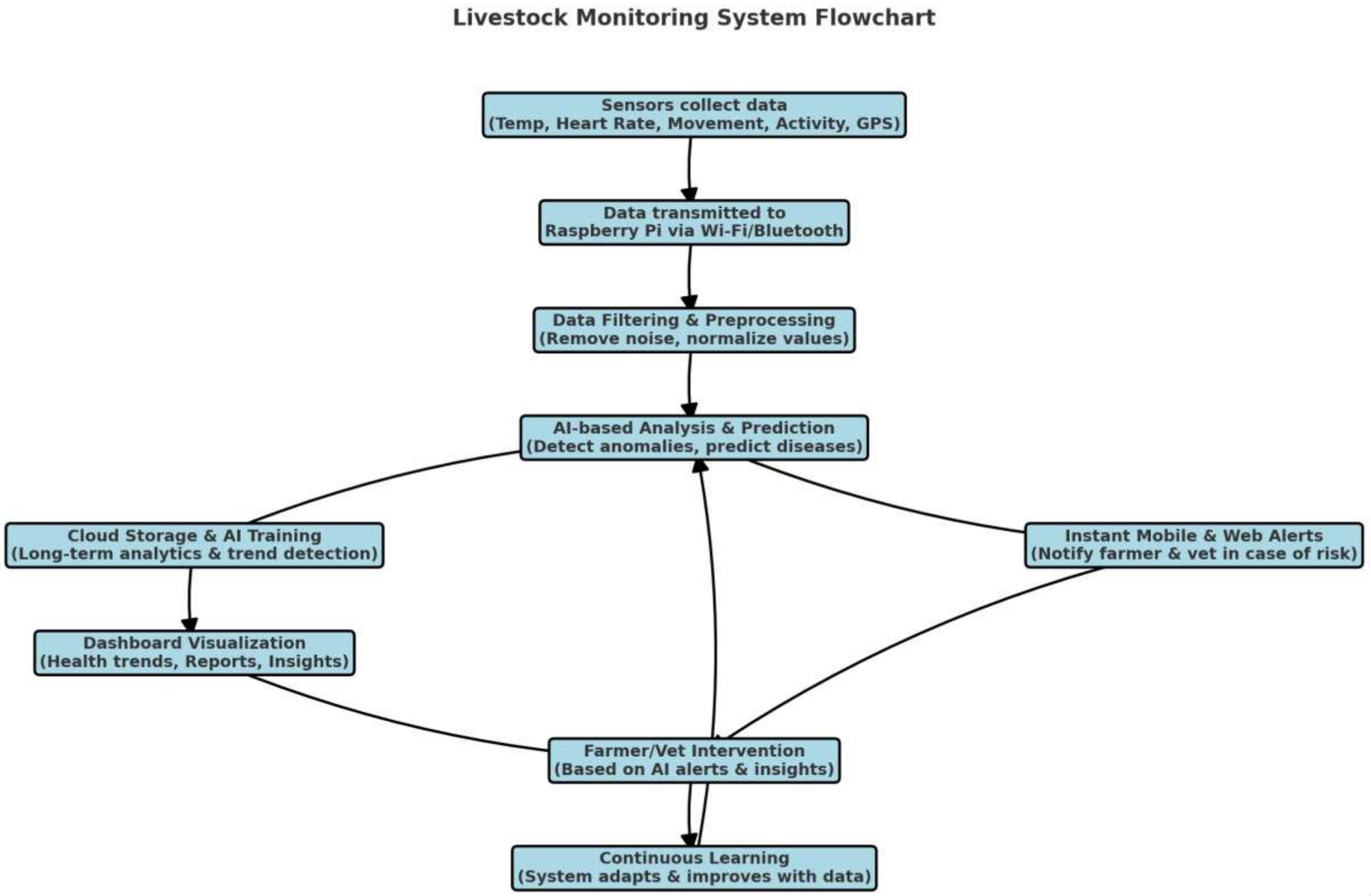
Credit: Data taken from WOAH (World Organization for Animal and Health)

PROPOSED SOLUTION

A real-time livestock health monitoring system using IoT, AI, smart sensors to enable early disease detection, reduce mortality rate and minimize economic losses.

- **Smart Wearables** track vital signs such as temperature, heart rate & blood flow.
- **AI Algorithms** predict disease outbreaks using collected data.
- **Mobile Alerts** notify farmers & veterinarians for early intervention.
- **Cloud Integration** allows for data-driven disease surveillance & analytics.

FLOWCHART


Livestock Monitoring System Flowchart
Sensors collect data
(Temp, Heart Rate, Movement, Activity, GPS)
Data transmitted to
Raspberry Pi via Wi-Fi/Bluetooth
Data Filtering & Preprocessing
(Remove noise, normalize values)
AI-based Analysis & Prediction
(Detect anomalies, predict diseases)
Cloud Storage & AI Training
(Long-term analytics & trend detection)
Instant Mobile & Web Alerts
(Notify farmer & vet in case of risk)
Dashboard Visualization
(Health trends, Reports, Insights)
Farmer/Vet Intervention
(Based on AI alerts & insights)
Continuous Learning
(System adapts & improves with data)

FLOWCHART EXPLANED

- **Sensors collect** real-time livestock health data (temperature, heart rate, movement).
- **Data is transmitted** via Wi-Fi/Bluetooth to Raspberry Pi for processing.
- **AI-based analysis** detects anomalies & predicts potential diseases.
- **Cloud storage** for long-term monitoring & analytics.
- **Mobile & Web alerts** notify farmers & veterinarians for early intervention.
- **Dashboard visualization** for user-friendly data access & insights.

FEATURES & NOVELTY

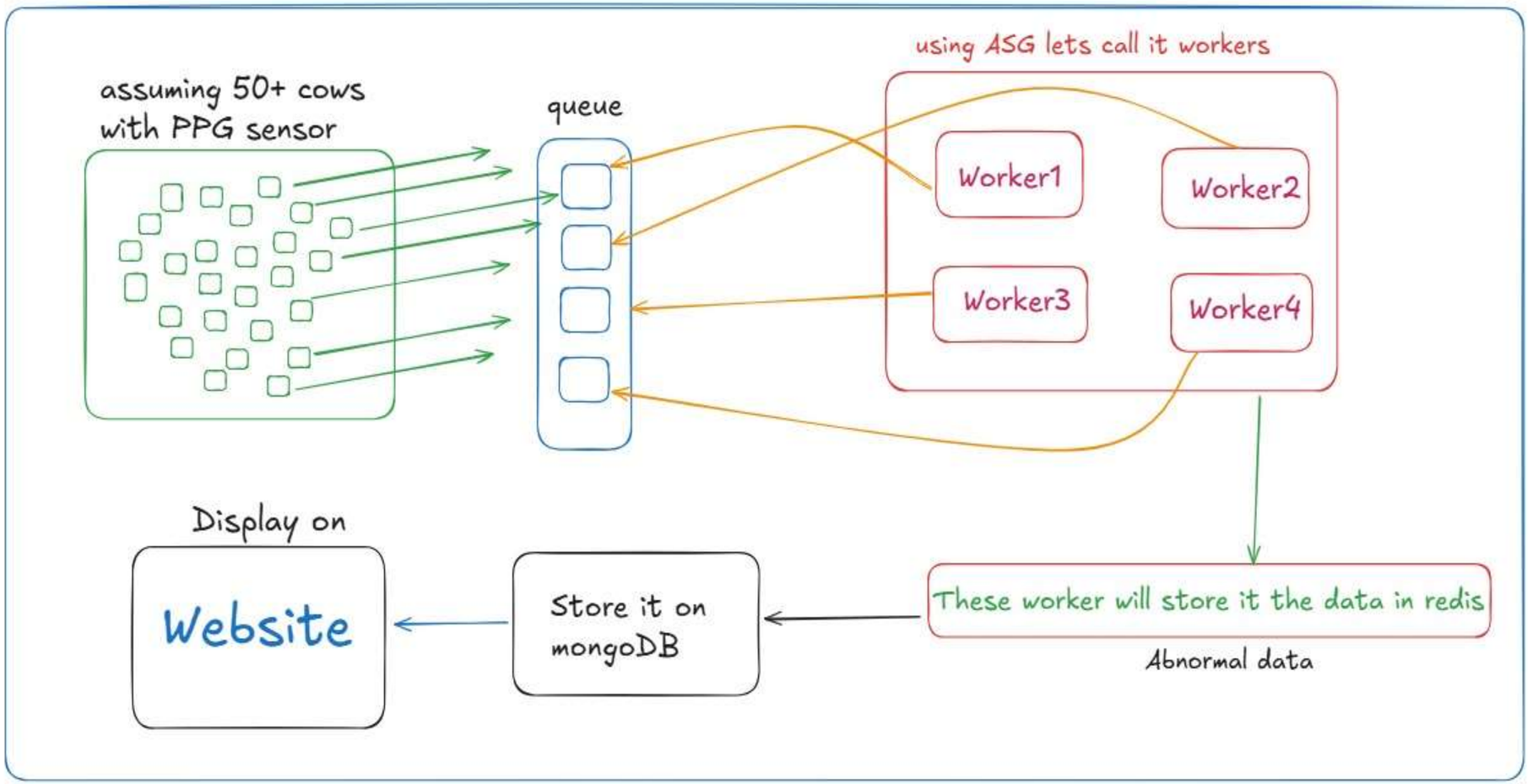
- **Real-Time Disease Detection** – Helps prevent outbreaks before they spread.
- **AI-Powered Health Insights** – Predicts illnesses early to reduce livestock mortality.
- **AI Chat Bot -** An AI chatbot that analyzes animal health data to deliver real-time insights.
- **Smart Wearables & IoT Sensors** – Enables continuous monitoring for precise health tracking.
- **Instant Alerts & Veterinary Support** – Quick treatment response minimizes losses.
- **Data-Driven Disease Surveillance** – Systematic collection, analysis, and interpretation of health-related data trends in real time.
- **Cost-Effective & Scalable** – Affordable for small farmers & large dairy industries.

CHALLENGES

- **Initial Setup Cost** – Hardware and IoT device costs may be a barrier for small-scale farmers.
- **Connectivity Issues** – Requires stable internet connectivity for real-time data transfer.
- **Data Privacy Concerns** – Secure storage and management of livestock health data are crucial.
- **User Adoption** – Training farmers to use digital solutions effectively.

SOLUTION

This make it scalable



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THANK YOU